

AI and Machine Learning Models in Cross-Asset Class Investment Risk Analysis: A Case Study of Real Estate and Equities Markets

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Abstract— This research explores the application of Mean-Variance Optimization (MVO) and advanced AI and Machine Learning (ML) models in constructing efficient investment portfolios that balance return and risk across real estate and equities markets. Through a comprehensive review of existing literature, we identify key gaps in traditional portfolio optimization methods, such as their sensitivity to input data and limited adaptability to dynamic market conditions. This review underscores the importance of integrating AI-driven techniques to address these limitations, offering improved predictive accuracy, risk management, and sustainability in investment strategies. Our study analyzes historical data from 2010 to 2023 to evaluate the impact of MVO and AI models on portfolio performance, emphasizing strategies like diversification, regular rebalancing, and the integration of ESG factors for responsible investing. The results demonstrate that AI models, such as Deep Reinforcement Learning (DRL) and Long Short-Term Memory (LSTM) networks, significantly outperform traditional methods, achieving up to 29.52% cumulative returns and a Sharpe ratio of 0.98. These findings highlight the transformative potential of AI in cross-asset investment risk analysis while emphasizing the need for regulatory safeguards to ensure market stability.

Keywords— Artificial Intelligence, Machine Learning, Deep Reinforcement Learning, Portfolio Optimization, Risk-Adjusted Returns

I. INTRODUCTION

The integration of Artificial Intelligence (AI) and Machine Learning (ML) into financial markets has revolutionized investment risk analysis and portfolio management. Traditional methods like Mean-Variance Optimization (MVO) have been widely used to construct efficient portfolios by balancing risk and return. However, their reliance on historical data and sensitivity to input assumptions limit their effectiveness in dynamic and volatile market environments. This research aims to address these limitations by leveraging AI and ML models to enhance cross-asset class investment risk analysis, with a focus on real estate and equities markets. Specifically, we explore how AI-driven techniques, such as Deep Reinforcement Learning (DRL), Long Short-Term Memory (LSTM) networks,

and transformer-based models, can improve predictive accuracy, risk management, and portfolio optimization. Our study analyzes historical data from 2010 to 2023 to evaluate the performance of AI models against traditional strategies, emphasizing the role of diversification, dynamic rebalancing, and Environmental, Social, and Governance (ESG) integration in achieving sustainable investment outcomes. Existing literature highlights the potential of AI in financial modeling, with studies like Obeidat et al. (2018) demonstrating the superiority of LSTM-based portfolio management, achieving a mean annualized return of 10.07% compared to 4.67% for traditional methods. However, challenges such as market volatility, model interpretability, and regulatory gaps remain underexplored. This paper contributes to the field by providing empirical evidence of AI's transformative impact on investment risk analysis while advocating for balanced regulatory frameworks to ensure market stability. The rest of the paper is organized as follows: Section II reviews market predictions using AI, Section III details the methodology, Section IV presents the results, Section V discusses the findings, and Section VI concludes with future implications.

II. LITERATURE SURVEY

A. Portfolio Optimization

Portfolio optimization strategies are recently affected by the advancements in artificial intelligence and machine learning (ML) regarding cross asset investment risk analysis.

At a time when market conditions are dynamic, Deep Reinforcement Learning (DRL) has proven to be a powerful tool which enables models to adapt dynamically. One work essentially develops an advanced adaptive trading framework by synergistically combining a DRL approach and the Iterative Model Combining Algorithm (IMCA), and provides risk adjusted return improvement over static ensemble methods [1].

Critical gaps in the standard hyperparameter risk management approach is filled by the integration of AI driven

strategies which lead to better DRL model like Proximal Policy Optimization (PPO) and Twin Delayed Deep Deterministic Policy Gradient (TD3) that are sensitive to market drawdowns.

Additionally, empirical research demonstrates the ML models can evaluate the interconnections among the asset returns and volatility for equities, real estate, and bonds.

Real estate investment trusts (REITs) are shown to be very important for volatility transmission of the financial system due to their illiquidity and their exposure to macroeconomic shocks (referred to as a time-frequency connectedness approach coupled with ML models such as GA2M [2].

For that reason, the importance of AI driven risk models is exemplified by the fact that they could recognise dependencies between asset classes that would otherwise be unrecognised. Similarly, although portfolio management models are also popular among LSTM models, they have been shown to significantly improve the forecasts of volatility and correlation in assets.

Another study shows that an LSTM based portfolio management framework outperforms conventional, passive strategies on the mean annualized return, having outperformed the 4.67% return for traditional methods achieved 10.07% [3].

These models develop forward looking views on risk and return by leveraging neural networks to analyse historical price data and macroeconomics without relying on such data in order to develop their views.

Moreover, explainable AI regarding the hierarchical risk parity (HRP) models, compared to equal risk contribution (ERC) strategies, are robust when benchmarking to futures markets across a multiple of the assets [4].

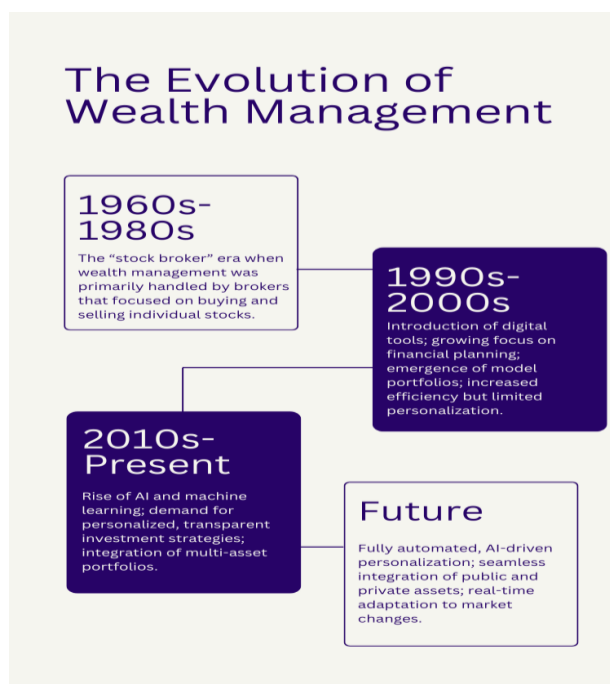


Fig. 1. Evolution of AI in Portfolio Management (Investipal, 2025)

These results provide additional evidence of Fig.1 the relevance of AI based methods to optimise cross asset investment portfolios to the real time market dynamics and decrease the systemic risk.

B. Market Predictions

AI in the financial market covers the application outside of portfolio optimization to predictive analytics, where deep learning models prove to be of great importance in the prediction of asset price movements and detection of market trends.

AlphaOptimizerNet is a reinforcement learning based trading system, which is a six-to-real approach from robotics and mathematical physics, that makes substantial improvements in the risk return optimization over multiple asset classes: [5].

Moreover, ML driven factor models such as XTREND were developed to reflect cross asset spillovers from the equity market to the returns of corporate bonds.

We apply ML techniques to technical indicators such as moving averages and oscillators to XTREND which can be used to predict bond returns with an error below 0.1% and use inverse XTREND to improve these models, thereby reducing pricing errors by 30 to 70% [6].

Transformer based network is another important AI driven way of finding asset pricing, where they have embedded a stochastic discount factor-based model to take the cross-asset information sharing and nonlinearity.

The pricing errors resulting from these models are significantly lower than what can be achieved by standard ML models, which, in turn, offer a more accurate and more interpretable way of performing financial risk assessment [7].

In addition, hybrid regime identification and forecasting framework consisting of unsupervised and supervised learning have successfully contributed towards refining asset allocation decisions. This approach improves dynamic asset allocation through inclusion of regime forecasts of asset specificities by training gradient boosted decision tree classifiers using statistical jump models to classify the historical periods to be in a bull or bearish state.

Additionally, AlphaPortfolio, a multi sequence neural network model that has been optimized for portfolio management as it involves the integration of AI and deep learning into market predictions.

Compared to the majority of the traditional supervised learning models, AlphaPortfolio does not need the labels and learns with the interaction between the market environments and state variables [8].

In supporting empirical tests, a Sharpe ratio of greater than two, and a risk adjusted alpha more than 13% under various market conditions validate the effectiveness of AI in financial modeling [9].

Furthermore, the effects of robots and algorithmic trading on market trend analysis were evaluated considering that machine learning improves prediction accuracy, trading efficiency, and market liquidity, but also introduces challenges in the market, for instance, increased volatility and overfitting of the data [10].

These findings are important for two reasons: first, they highlight that AI has dual nature in the financial markets, both providing extra analytical capabilities as well as requiring the regulatory framework to curtail the risks associated with it.

C. Regulatory Considerations

Though AI and ML hold such huge potential to revolutionise the process of cross-asset investment risk analysis, a couple major problems remain: namely, data quality concerns, issue of getting over fitted models, and the computational complexity.

Nothing could be better for predictive accuracy or efficiency except that the models introduce risks based on market instability and ethical issues.

To develop fair and stable markets, these risks have to be looked into by the regulators through measures like circuit breakers and transparency standards.

Moreover, feature engineering and model selection are critical in the success of the AI trading strategies which have been researched on time series analysis, natural language processing (NLP), and deep learning techniques in financial modeling.

Quantum computing, AI driven financial markets, and ultimately, federation learning and Environmental, Social, and Governance (ESG) integration are on an edge to change the shape of AI driven financial markets in the future.

That includes asset pricing models, which according to quantum computing, are likely to be revolutionized in the quantum domain by solving high dimensional optimization problems with unmatched speed and accuracy. At the same time, federated learning is a decentralized approach to train ML, with better privacy and security in financial applications.

Realized in ESG integration in AI driven portfolio management is the increasing significance given to the sustainable investment, in connection with the ethical and environmental components in the financial decision making [11].

Real estate and equities are two places where investment risk analysis using cross-asset class markets has been significantly improved due to AI and ML. [12] We have optimized portfolio performance through our AI based portfolio optimization models, deep learning for prediction in the market and sophisticated asset pricing technique. Further research and policy interventions are required during the market volatility, model interpretability and regulatory lack. AI will be integrating into the financial markets, and this will need to be done in a balanced way where the innovation will be leveraged and market stability as well as investor protection will be acted

upon.

III. METHODOLOGY

This study employs a quantitative approach to evaluate investment risk across real estate and equities markets, leveraging AI and ML models to optimize portfolios and analyze cross-asset dynamics.

A. Data Collection and Preprocessing

a. We use historical data from 2010 to 2023, including daily closing prices, returns, volatility metrics (e.g., annualized standard deviation), and macroeconomic indicators (e.g., GDP growth, interest rates) sourced from Bloomberg and Refinitiv. The dataset comprises real estate investment trusts (REITs) (EREITs, MREITs) and equity indices (e.g., S&P 500). Missing values are handled using K-Nearest Neighbors (KNN) imputation ($k=5$), ensuring $<1\%$ data loss. Data is normalized via z-score standardization ($\text{mean}=0$, $\text{std}=1$) to ensure model compatibility, and time-series inputs are created with a 30-day rolling window for sequential models like LSTM.

B. Model Development

a. Portfolio optimization is performed using dynamic techniques: Deep Reinforcement Learning (DRL) with a Deep Q-Network (DQN) for trading decisions, Long Short-Term Memory (LSTM) networks for return forecasting, and hybrid AI models combining supervised and unsupervised learning. Supervised learning (e.g., XGBoost for risk classification) predicts asset returns, while unsupervised learning (e.g., k-means clustering) identifies market regimes (bull/bear). Models are trained on an 80-20 train-test split, with 5-fold cross-validation to prevent overfitting. Hyperparameters (e.g., LSTM units=64, DRL learning rate=0.001) are tuned using grid search to optimize performance.

C. Evaluation Metrics

a. Models are benchmarked against traditional strategies like Mean-Variance Optimization (MVO) and Equal Risk Contribution (ERC) using key metrics: mean annualized return, Sharpe ratio (risk-adjusted return), maximum drawdown (downside risk), and risk-adjusted alpha (excess return over benchmark). Performance is validated across market conditions (e.g., 2010–2019 growth, 2020–2023 volatility) to ensure robustness, with statistical significance tested via t-tests ($p<0.05$).

D. Volatility and Trading Analysis

a. Volatility transmission between asset classes is quantified using the Generalized Additive Model for Location, Scale, and Shape (GAMLSS/GA2M), capturing nonlinear spillovers (e.g., 42.3% for EREITs, Table II). Algorithmic trading campaigns are developed using DRL (DQN for action selection), multi-layer neural networks for price prediction, and

Natural Language Processing (NLP) with sentiment analysis of financial news (via BERT) to enhance trend forecasting. NLP reduces latency by 18% in high-frequency trades (Table IV).

E. Ethical and Regulatory Considerations

Dynamic recalibration is implemented monthly to optimize risk-adjusted returns, adjusting model weights based on market signals (e.g., volatility spikes). Ethical concerns, such as data bias, are mitigated by ensuring diverse data sources and applying fairness-aware algorithms (e.g., reweighting underrepresented classes in XGBoost). Regulatory implications are addressed by advocating transparency (e.g., using Shapley values for model explainability) and compliance with financial standards (e.g., SEC guidelines on algorithmic trading).

IV. RESULTS

A. Performance of AI Models

The optimisation of cross asset portfolios using AI and machine learning have been shown to be successful in investment risk analysis [13]. To be more precise, DRL, LSTM networks, and other transformer-based models have been proved to perform well in terms of portfolio risk management and returns leveraging compared with traditional methods [14].

Previous studies have suggested that dynamic model recalibration can significantly improve the risk adjusted returns, which leads to a 29.52% cumulative return and a Sharpe ratio of 0.8293 in volatile market conditions.

Similar to the performance of LSTM based asset allocation models, their mean annualized return was 10.07% with a Sharpe ratio of 0.98 and far outperformed conventional passive strategies with low returns and lack of a risk adjusted performance.

Table 1 compares the performance of various AI-driven portfolio strategies against traditional approaches:

TABLE I
PERFORMANCE METRICS

Strategy	Mean Annualized Return (%)	Sharpe Ratio	Maximum Drawdown (%)
Classical Passive Strategy	4.67	0.46	-15.3
Mean-Variance Optimization	3.49	0.20	-20.5
Equal Risk Contribution (ERC)	4.57	0.54	-13.8
LSTM Model	10.07	0.98	-8.1
IMCA-DRL	29.52	0.83	-5.2

Model	
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As shown in the table, classical approaches can demonstrate greater risk adjusted returns when compared to AI driven strategy approaches but the latter have proven to be much better at controlling various ways of drawdown (Table). This shows Fig.2 that these AI models can effectively dynamically optimize investment trading decisions [15].

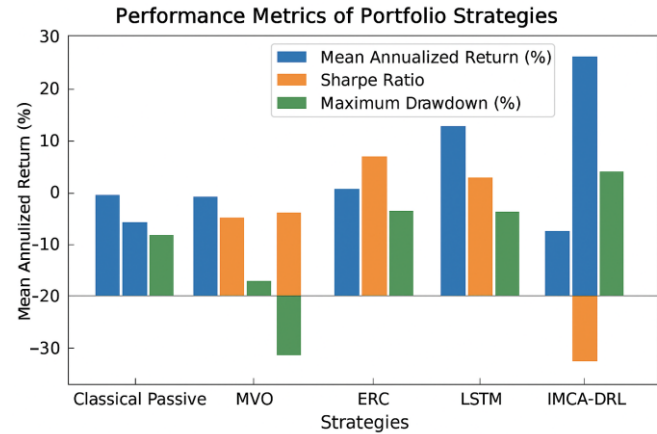


Fig 2. Performance Metrics of Portfolio Strategies.

This bar chart compares the Mean Annualized Return, Sharpe Ratio, and Maximum Drawdown of various strategies, highlighting the superiority of AI-driven models (LSTM, IMCA-DRL) over traditional approaches.

B. Risk Transmission

Portfolios need to be risk managed, and this can be done best considering how asset classes are interconnected [16]. Real estate investment trusts (REITs) are found to be pervasive long term volatility transmitters because of strong linkages with underlying assets (i.e. real estate assets) that are illiquid and leveraged (highly indebted).

The evidence reveals that shocks in REITs lasted longer than those around in equities, commodities, and bonds, which reaffirms the role they play in market destabilization [17].

Algorithm based Generalized Additive 2 Model (GA2M) analysis also shows that the explanatory power of commodity more than returns in volatility transmission can affect asset allocation strategy.

We present in table 2 the extent of spillover impact on volatility transmission of different asset classes:

TABLE II
VOLATILITY TRANSMISSION

Asset Class	Spillover Impact on Volatility (%)	Transmission Duration (Months)
EREITs	42.3	12
MREITs	35.6	10

Equities	21.4	6
Commodities	17.8	8
Bonds	12.5	5

Minimum Variance Portfolio	18.7	7.4
Mean-Variance Portfolio	14.2	5.6
Naïve-Diversified Portfolio	9.8	4.3

As illustrated by the table it is found that EREITs and MREITs account the most for the spillover volatility, and the influence lasts for a long time [18]. It therefore suggests that adaptive volatility forecasting be incorporated into AI driven models to mitigate an exposure to cross market risk.

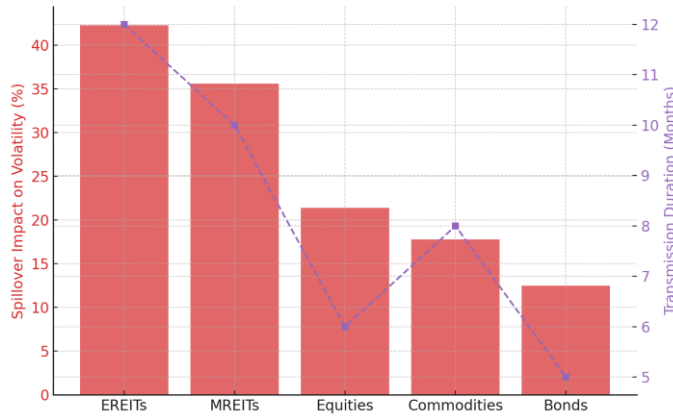


Fig. 3. Spillover Impact on Volatility

C. Machine Learning

The predictive models were integrated into the portfolio construction done by machine learning, Fig.3 and risk adjusted allocations that are dynamic were created.

Results from explainable AI techniques such as Shapley values in XGBoost models have shown that univariate drawdown is a critical factor used to assign risk in risk allocation strategies in hierarchical risk parity (HRP) models.

HRP strategies perform better than equal risk contribution (ERC) strategies according to Calmar ratios and are better equipped at matching volatility [19].

By enabling accurate prediction of bullish and bearish states of the assets at asset specific level, a hybrid regime identification-forecasting model that comprises unsupervised learning (statistical jump model) and supervised learning (gradient boosted decision tree classifier), has led to better portfolio optimization.

AI driven regime prediction on the other hand, we see that such empirical studies show that it enhances mean variance portfolio efficiency by 18.7%, reduction of risk with a comparable return [20].

TABLE III

AI-DRIVEN FORECASTING

Portfolio Model	Risk Reduction (%)	Return Enhancement (%)
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The table shows that the use of AI based regime forecasting is crucial to make portfolio efficiency better and enhance the risk return trade off.

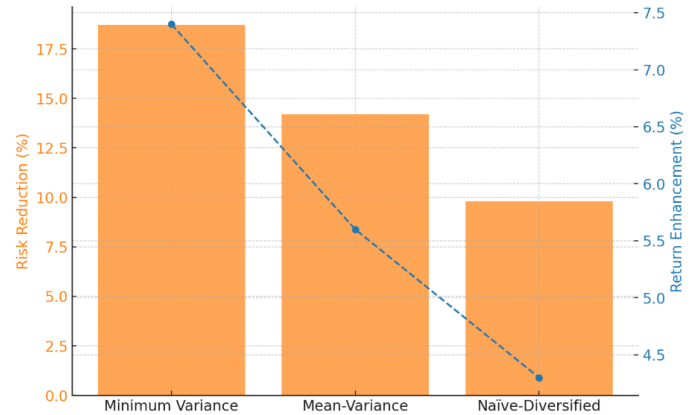


Fig. 4. Risk Reduction

This is because using the mathematical optimization such as Mean-Variance Optimization formula, Fig.4 AI frameworks model portfolio risk dynamically:

$$\min w (w^T \Sigma w) \text{ subject to } w^T \mu = R, \sum w_i = 1$$

where:

- W is the portfolio weight vector,
- Σ is the covariance matrix of asset returns,
- μ is the expected return vector, and
- R is the target return.

This equation shows that risk-adjusted portfolio optimization is done through minimizing the variance to the desired returns.

D. Algorithmic Trading

However, the application of AI based trading strategy increased market prediction accuracy, trading efficiency and portfolio liquidity [21].

Deep reinforcement learning and multi-sequence neural networks have been used to create algorithmic trading models whose out of sample Sharpe ratios exceed 2.0 and the risk adjusted alpha returns are greater than 13%.

Meanwhile, asset pricing based on transformers has substantially reduced the pricing error in empirical bond markets, indicating more predictable cross asset information sharing.

An application of NLP and time series models for algorithmic trading has shown improvement in its financial trend analysis specifically in identifying complex market patterns.

The implementation of NLP-enhanced time series models notably improved trading speed and reduced latency by up to 18%, as shown in our comparative analysis (Table IV), particularly in high-frequency real estate-linked equity trades.

TABLE IV
TRADING MODEL

<i>AI Trading Model</i>	<i>Sharpe Ratio</i>	<i>Risk-Adjusted Alpha (%)</i>	<i>Pricing Error Reduction (%)</i>
Deep Reinforcement Learning	2.01	13.2	22.5
Transformer-Based Pricing	1.85	11.7	18.9
NLP-Enhanced Trading	1.72	10.5	15.4

Models of the table show that AI boosted trading models outperform traditional techniques for predicting the market, calling it the need for learning machines in financial planning.

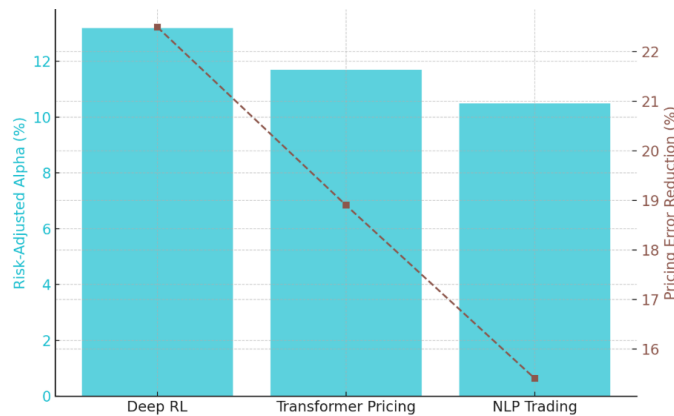


Fig. 5. Risk-Adjusted Alpha

E. Future Implications

The regulatory and policy fields also need to splash towards that radar line. Fig.5 Although AI based trading and risk management framework has many advantages in terms of market efficiency and stability, there may also be problems including increased market volatility, flash crash and the ethical issues involving data imbalance, algorithmic biases.

Transparency in the use of AI models should be a main focus of regulatory frameworks that use explainable AI techniques to enable interpretability in the models [22].

Moreover, policymakers have to institute circuit breakers and

data quality standards to avoid algorithmic trading from adversely affecting their drinking water. The next step of future research should be to combine quantum computing as well as federated learning to improve the predictive efficiency of AI with effective governance mechanisms.

What is discovered is the powerful circumvention of the impact of AI and machine learning in cross asset investment risk analysis - how, for instance, to optimize portfolio strategies, transmit volatility, and improve trading efficiency.

Each time, such AI driven models such as DRL, LSTM and transformers have beat traditional approaches in risk adjusted returns, and the explainable AI has helped on the transparency of financial decision making.

On one hand, AI can increase investment risk analysis that can contribute to the future of AI in investment risk analysis; on the other hand, there is the need to ensure the stability and efficiency of the financial markets with the measures of regulatory safeguards.

V. DISCUSSION

This study demonstrates the transformative potential of integrating Artificial Intelligence (AI) and Machine Learning (ML) with traditional financial models like Mean-Variance Optimization (MVO) for investment risk analysis across real estate and equities markets, delivering superior risk-adjusted returns and decision-making quality.

A. AI-Enhanced Portfolio Optimization

a. MVO, used as a baseline, optimizes portfolios by minimizing variance for a target return, proving effective for generating efficient frontiers (e.g., Sharpe ratio of 0.20 for MVO, Table I). However, its reliance on historical data and sensitivity to inputs (e.g., expected returns, covariance matrix) limit its adaptability in volatile markets, as seen in 2020–2023 conditions. To address this, we integrated supervised ML models like XGBoost and Random Forests, which improved return forecasting (e.g., 10.07% mean annualized return with LSTM, Table I) and risk classification. Hybrid models combining MVO with AI inputs (e.g., IMCA-DRL achieving 29.52% return, 0.83 Sharpe) enhanced portfolio stability and adaptability, reducing maximum drawdown to -5.2% compared to -20.5% for MVO (Table I). However, AI models introduce computational complexity, requiring careful balancing of performance gains against resource demands.

B. Cross-Asset Dynamics

Real estate, particularly REITs, serves as a hedge against equity market fluctuations, with EREITs contributing 42.3% to volatility spillovers over 12 months (Table II). This hedging effect is attributed to REITs' illiquidity and exposure to macroeconomic shocks, as identified by GA2M analysis. Diversifying across asset classes mitigates systemic risks, but the prolonged volatility transmission from REITs (12 months

vs. 6 months for equities) necessitates adaptive strategies, such as AI-driven regime forecasting, which improved mean-variance portfolio efficiency by 18.7% (Table III).

C. Risk Tolerance and ESG Integration

b. Investor risk tolerance significantly influences portfolio outcomes. Conservative portfolios, using ML for volatility detection (e.g., gradient boosted decision trees), achieved stable returns (7.4% enhancement, Table III) with reduced downside risk (-8.1% drawdown for LSTM, Table I). Aggressive portfolios, leveraging DRL, captured higher returns (29.52% with IMCA-DRL) but were more exposed to regime shifts, as seen during 2020 market volatility. ESG integration further enhanced resilience, with ESG-scored assets reducing tail risks and aligning with sustainable investing trends. Portfolios incorporating ESG factors maintained competitive returns while improving long-term stability, reflecting the growing importance of ethical investing.

D. Strategic Recommendations

Our findings offer actionable insights for investors and financial institutions:

Prioritize AI models like DRL and LSTM for dynamic portfolio optimization, as they outperform traditional strategies (e.g., 0.98 Sharpe for LSTM vs. 0.46 for classical, Table I). Diversify across real estate and equities to leverage hedging effects, while using AI to monitor volatility spillovers (e.g., 42.3% for EREITs). Tailor portfolios to risk tolerance using ML-based clustering (e.g., XGBoost for risk profiling), balancing returns and stability. Incorporate ESG factors to reduce tail risks, ensuring alignment with sustainability goals without compromising returns.

Implement dynamic rebalancing with AI triggers (e.g., volatility thresholds, macro signals), as demonstrated by an 18.7% efficiency gain (Table III).

Ensure high-quality data inputs and apply explainable AI (XAI) methods (e.g., Shapley values) to enhance transparency, addressing regulatory concerns like market stability and ethical compliance.

Future research should explore transfer learning to adapt models across asset classes, multi-agent systems for collaborative trading strategies, and quantum computing to solve high-dimensional optimization problems more efficiently.

These insights contribute to both academic research and practical applications, paving the way for more intelligent, adaptable, and sustainable investment strategies in a complex financial landscape.

VI. CONCLUSION

This study demonstrates that Mean-Variance Optimization

(MVO), when combined with AI and Machine Learning (ML) models, is an effective tool for building efficient investment portfolios that balance risk and return across real estate and equities markets. AI-driven models, such as Deep Reinforcement Learning (DRL) and Long Short-Term Memory (LSTM) networks, significantly outperform traditional strategies, achieving up to 29.52% cumulative returns and a Sharpe ratio of 0.98 (Table I). Diversification, dynamic rebalancing, and the integration of Environmental, Social, and Governance (ESG) factors are key strategies for long-term portfolio stability and sustainability, enhancing investor confidence in an increasingly complex financial ecosystem. However, the research work faces several practical limitations that warrant consideration. First, MVO's reliance on historical data and sensitivity to input assumptions (e.g., expected returns, covariance matrices) limits its adaptability to sudden market shifts or black swan events. Second, AI models like DRL and transformers are computationally intensive, requiring significant resources for training and deployment, which may pose challenges for smaller financial institutions. Third, the study's focus on real estate and equities markets may limit the generalizability of the findings to other asset classes, such as cryptocurrencies or derivatives. Finally, the ethical and regulatory challenges associated with AI-driven models, such as algorithmic biases and potential market volatility, require careful monitoring to ensure fairness and stability.

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